

TOPICAL REVIEW

Security Issues in Multi-Cloud: A Systematic Literature Review

**DRAJAT CAHYA DININGRAT^{ID}, SUHARDI, (Member, IEEE),
AND BUDI RAHARDJO, (Member, IEEE)**

School of Electrical Engineering and Informatics, Bandung Institute of Technology, Bandung 40132, Indonesia

Corresponding author: Drajat Cahya Diningrat (23223012@mahasiswa.itb.ac.id)

ABSTRACT This study aims to systematically review the current literature regarding security concerns in multi-cloud environments. The main objective is to identify critical security issues, current solutions, investigate challenges, and propose recommendations for future research. This study uses a systematic literature review methodology adopting the Kitchenham method, which includes a literature search across leading academic databases including IEEE Xplore, ScienceDirect, ACM, SpringerLink, Emerald Insight, Nature, Taylor & Francis, Scopus, and ProQuest. A formal protocol was followed to conduct an automated search of relevant articles published between 2021 and 2024, and filters were applied. Based on the initial search results, 2038 studies were found, then the selection criteria process (inclusion and exclusion) and quality assessment were carried out, and finally 30 studies were selected for further research. The results show that most of the multi-cloud security issue trends fall into the areas of data, cryptography, authentication and authorization, followed by areas of service availability, confidential computing, and non even fall into the area of virtualization. In addition, this study systematically analyzes the relevant literature and identifies problems, solutions, challenges, and recommendations for future research. The findings of this review are expected to provide a strong overall picture for further research in multi-cloud environments.

INDEX TERMS Multi-cloud, security, multi-cloud security.

I. INTRODUCTION

In recent years, organizations have increasingly favored multi-cloud adoption because of its benefits in terms of flexibility, availability, and reduction of single-provider failure risk. Nevertheless, multi-cloud environments present considerable challenges regarding security, interoperability, and data management that single-cloud architectures do not adequately address. The security challenges in multi-cloud environments arise from the variability of security policies and standards among providers, resulting in configuration inconsistencies, complex authorization processes, and heightened risks of security breaches. Research indicates that the primary challenges in multi-cloud environments encompass safeguarding data integrity, authentication, and managing access rights due to the diversity of cloud providers. The study offers insights into preserving data integrity in cloud storage, pertinent to the realm of multi-cloud security.

The associate editor coordinating the review of this manuscript and approving it for publication was Amjad Mehmood^{ID}.

Simultaneously, the observed phenomenon pertains to the transition in cloud utilization from reliance on a singular provider to a consortium of providers. This is also demonstrated through research [1] that analyzes the cloud federation architectures and the functional as well as non-functional properties that influence how multiple cloud providers can collaborate.

Another challenge is interoperability among service providers, which frequently constitutes a significant obstacle for organizations seeking to fully exploit a multi-cloud strategy. The study [2] introduced the multi-cloud interoperability framework (MCIF) to facilitate seamless communication and data transfer among family providers via cloud ontologies and centralized identity management. The framework is especially pertinent for mitigating vendor lock-in, which complicates users' ability to transition between service providers without jeopardizing their security and performance.

As the adoption of multi-cloud technologies increases, security challenges related to data breaches, encryption, and strict access control become increasingly critical.

For example, the study [3] highlight key threats such as side-channel attacks and authorization attacks that can occur due to virtualization systems in multi-cloud environments. Case studies illustrating how minor configuration errors can lead to data breaches, as encountered by numerous large corporations in documented security incidents.

With the growing complexity of multi-cloud environments, there is a necessity for a more adaptive security strategy utilizing emerging technologies like machine learning and artificial intelligence. These technologies can be employed to proactively identify threats and deliver early warnings of potential cyberattacks. The study [4] focuses on bio-inspired algorithmic approaches for resource optimization in cloud/multi-cloud environments, which potential in optimizing resource allocation, scalability, fault tolerance, and security in cloud and multi-cloud. While these technologies are not directly targeted at threat detection, their ability to dynamically to changing system conditions is useful for developing robust and secure infrastructure.

The study [5] discusses the need for an effective trust management approach in multi-cloud environments, particularly through the adoption of a zero trust architecture that emphasizes access verification at every layer of interaction between systems. This approach is relevant to the data security method proposed in the study [6], which surveys the challenges faced by organizations seeking to ensure privacy, policy, data integrity, data encryption, and data partitioning in multi-cloud environments.

Several previous review studies [1], [2], [3], [4], [5], [6] have provided various perspectives ranging from functional and non-functional properties, then there are those that focus on interoperability, then security issues, bio-inspired algorithmic framework, trust management framework, and data security in multi-cloud. So that there is no systematic literature review that comprehensively provides an overview of current security issue trends, problems, currently proposed solutions, and challenges, as well as recommendations for future research directions.

This systematic literature review will examine four research inquiries: (1) security concerns in multi-cloud environments; (2) existing solutions, or strategies, or methodologies for mitigating multi-cloud security issues; (3) challenges associated with multi-cloud security; (4) suggestions for future research directions in multi-cloud security. This systematic literature review primarily aims to deliver an in-depth comprehension of multi-cloud security problems, existing solutions or strategies, challenges, and avenues for future research. It will provide insights into the present and forthcoming trends in multi-cloud security domains that have experienced substantial advancements.

The structure of this paper is as follows: Section I contains the introduction, Section II is the research method used, and contains the Systematic Literature Review (SLR) stages based on Kitchenham Guidelines. Section III presents the results and discussion related to trend security issues, problems, solutions, challenges, and recommendations. Finally,

Section IV contains the research conclusions and suggestions for further systematic literature review research.

II. RESEARCH METHOD

This research employs a systematic literature review methodology grounded in Kitchenham's recommendations. The recommendations comprise three stages, detailed as follows:

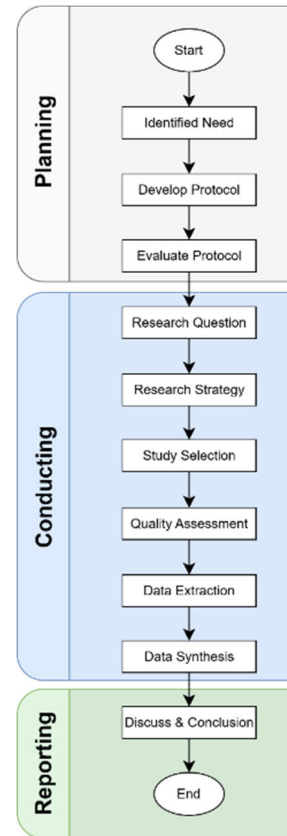


FIGURE 1. Phases of a systematic literature review based on Kitchenham [7].

Planning Stage: It is imperative to ascertain the necessity of a systematic literature review prior to its commencement. At this juncture, the objectives are to (a) acknowledge the need for a review, (b) plan the research questions, (c) develop the research protocol and (d) evaluate the review protocol [7].

Conducting stage [7], includes several steps, namely:

- **Research Question,** The objective of the RQ stage is to direct the research focus and constrain the scope to ensure relevance to the topic of multi-cloud security.
- **Research Strategy,** The principal objective of performing a systematic review is to locate the maximum amount of original research pertinent to a specific research question. This was accomplished by employing an impartial and equitable search strategy. The degree of rigor in the search process is a distinguishing factor between systematic and traditional reviews.
- **Study Selection,** The objective of formulating study selection criteria is to discern the most pertinent studies

that can furnish direct evidence for the research inquiry. To mitigate bias, these selection criteria should be delineated at the onset of the research protocol development, although they may be adjusted during the information retrieval phase. The criteria for inclusion and exclusion must align with the research question. In this process, included studies must be evaluated to ensure consistent interpretation and accurate classification.

- **Quality Assessment**, This phase seeks to evaluate the methodological rigor and dependability of the studies incorporated in the chosen literature review. This is a crucial measure for preserving the integrity and validity of the SLR outcomes. It also guarantees that the literature review concentrates on high-caliber studies pertinent to the research objectives.
- **Data Extraction**, create data collection forms to precisely document information acquired by researchers from primary studies. To mitigate potential bias, data collection instruments should be developed and validated during the formulation of the research protocol.
- **Data synthesis**, systematize and articulate the results of the chosen studies succinctly. This procedure entails a sequence of steps that must be meticulously adhered to in order to improve comprehension of the synthesized data.

Reporting Phase, this stage represents the conclusive phase of the systematic review process, involving the documentation and dissemination of the review's findings to relevant stakeholders [7].

A. RESEARCH QUESTIONS

The formulation of the research questions is a paramount aspect of any systematic study [7]. Research inquiries were developed utilizing the PICOC (Population, Intervention, Comparison, Outcome, Context) framework. The PICOC for this systematic literature review are presented in Table 1. The subsequent text provides an elucidation of PICOC.

TABLE 1. Overview of PICOC.

PICOC	Description
Population	Studies and literature that discuss security in multi-cloud.
Intervention	Security solutions, strategies and practices implemented in a multi-cloud environment
Comparison	N/A
Outcomes	Identification of key security issues, current solution, research challenges, and recommendations for future research
Context	Multi-cloud environments used by different types of organizations, including factors such as company size, industry, and applicable regulations.

The population in this study encompasses all the research and literature addressing security in multi-cloud

environments. This encompasses diverse publications that emphasize security considerations in multi-cloud environments, including technical papers, scientific studies, and conference articles that elucidate the management of security within intricate multi-cloud ecosystems.

The interventions addressed pertain to security solutions, strategies, and practices employed in multi-cloud environments. These solutions encompass the technical measures and policies employed to safeguard the data, applications, and infrastructure distributed across various cloud service providers. The emphasis is on comprehending the strategies employed to alleviate security threats particularly in multi-cloud environments.

This study lacks a comparative element, as the objective of this systematic literature review is to identify security issues and solutions specifically within the multi-cloud context, without juxtaposing them with other architectures such as single cloud or hybrid cloud. The objective is to examine and condense findings from the pertinent literature.

This study aims to identify critical security issues in multi-cloud environments, examine implemented solutions to address these challenges, highlight unresolved research problems, and provide recommendations for future research endeavors. These findings are anticipated to enhance the literature by providing a valuable summary for practitioners and scholars engaged in the advancement of multi-cloud security solutions.

This study pertains to a multi-cloud environment utilized by diverse organizations, from small enterprises to large corporations, across multiple industries. This context also considers external factors such as organizational size, industry sector, and relevant regulations, which can influence the implementation of security challenges and solutions in a multi-cloud environment.

The research questions were derived from the PICOC framework as follows:

- What are the primary security issues faced by organizations adopting multi-cloud strategies?
- What solutions, or strategies, or security practices have been implemented to mitigate security issues in multi-cloud environments?
- What are the main research challenges associated with addressing security issues in multi-cloud environments?
- What recommendations can be made for future research to enhance security in multi-cloud environments?

B. SEARCH STRATEGY

The search process entails selecting digital libraries, establishing search keywords, executing the search, and acquiring a preliminary list of pertinent primary studies. The search strategy aids in formulating the suitable search string and selecting relevant databases to gather pertinent documents for addressing the research question.

A method to identify high-quality literature sources is to utilize a specialized library database within the discipline. Literature searches are performed in prominent digital libraries within this domain to identify diverse literature studies. This study utilizes the following digital libraries:

- 1) ACM Digital Library (<https://dl.acm.org/>)
- 2) Emerald Insight (<https://www.emerald.com/insight/>)
- 3) IEEE Xplore (<https://ieeexplore.ieee.org/>)
- 4) Nature (<https://www.nature.com/>)
- 5) ProQuest (<https://www.proquest.com/>)
- 6) ScienceDirect (<https://www.sciencedirect.com/>)
- 7) Scopus (<https://www.scopus.com/home.uri>)
- 8) SpringerLink (<https://link.springer.com/>)
- 9) Taylor & Francis (<https://www.tandfonline.com/>)

A strategy for conducting searches to identify papers for analysis in this systematic literature review can be developed based on the principal digital libraries. The primary constraint is that the quantity of accessible databases is significantly influenced by the search terms and filters applied. The filters applied consist of publication years from 2021 to 2024, and the publication type is restricted to technical papers, including journals as specified in the selected database search parameters. Nonetheless, the terminology or nomenclature for publication filters varies across databases, yet their function remains consistent.

The development of search terms entails multiple steps: (1) extracting search terms from PICOC that concentrate on populations and interventions, (2) deriving search terms from research topics and inquiries, and (3) locating synonyms for the identified terms. Subsequently, the refinement of search terms is achieved by constructing more intricate search queries utilizing the identified terms, alongside the application of Boolean logic operations ‘AND’ and ‘OR’. The search series employed in the investigation is presented in Table 2.

The preliminary search strategy employed uniform keywords across all databases: (“multi-cloud” OR “multi cloud”) AND (“security” OR “cybersecurity” OR “secure*”), noting that the ScienceDirect database explicitly prohibits the use of wildcards. Subsequently, filters for year and others were applied, yielding preliminary findings that had not yet been established as references in the discussion. The initial search yielded a total of 2038 papers, with the specifics of the papers retrieved from each database presented in Figure 2.

C. STUDY SELECTION CRITERIA

At this juncture, inclusion and exclusion criteria were established to mitigate bias and concentrate on the research subject. The criteria for inclusion and exclusion in this systematic literature review are presented in Table 3.

Figure 3 delineates the comprehensive search procedures and the quantity of studies identified at each stage. The Mendeley software and Parsifal tool were utilized to manage the search results. The systematic literature reviews report section reference guide was utilized during the study selection

TABLE 2. Keyword Search and filter.

Database	Keyword	Filter
ACM Digital Library	("multi-cloud" OR "multi cloud") AND ("security" OR "cybersecurity" OR "secure*")	2021-2024 AND Research Paper AND Journal
Emerald Insight	("multi-cloud" OR "multi cloud") AND ("security" OR "cybersecurity" OR "secure*")	2021-2024 AND Article
IEEE Xplore	("multi-cloud" OR "multi cloud") AND ("security" OR "cybersecurity" OR "secure*")	2021-2024 AND Journal
Nature	("multi-cloud" OR "multi cloud") AND ("security" OR "cybersecurity" OR "secure*")	2021-2024 AND Research
ProQuest	("multi-cloud" OR "multi cloud") AND ("security" OR "cybersecurity" OR "secure*")	2021-2024 AND Scholarly Journal
ScienceDirect	("multi-cloud" OR "multi cloud") AND ("security" OR "cybersecurity" OR "secure*")	2021-2024 AND Research Articles
Scopus	("multi-cloud" OR "multi cloud") AND ("security" OR "cybersecurity" OR "secure*")	2021-2024 AND Article AND Journal AND English
SpringerLink	("multi-cloud" OR "multi cloud") AND ("security" OR "cybersecurity" OR "secure*")	2021-2024 AND Research Article AND English
Taylor & Francis	("multi-cloud" OR "multi cloud") AND ("security" OR "cybersecurity" OR "secure*")	2021-2024 AND Article

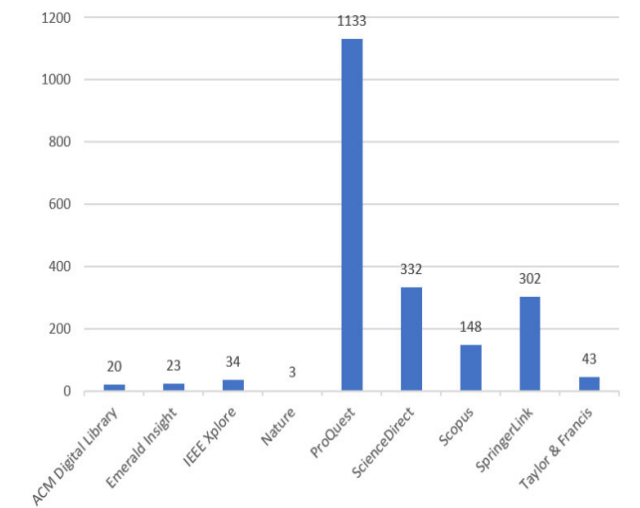
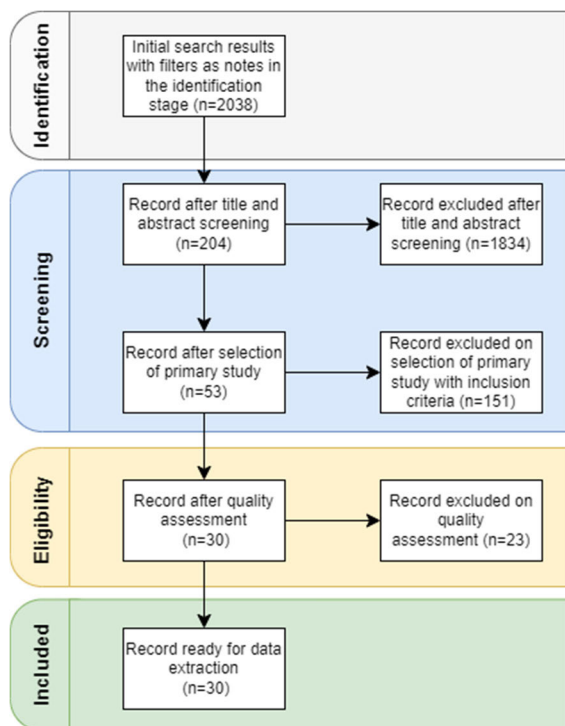


FIGURE 2. Initial search results with filters.

process, with the inclusion and exclusion criteria elaborated upon in Table 3. Figure 3 illustrates that the process was segmented into multiple stages: initially, an exploratory search was performed utilizing specified keywords and filters as outlined in Table 2, yielding 20 articles from the ACM Digital Library, 23 articles from the Emerald Insight database, 34 articles from the IEEE Xplore database, 3 articles from the Nature database, 1133 articles from the ProQuest database, 332 articles from the ScienceDirect database, 148 articles from the Scopus database, 302 articles from the SpringerLink database, and 43 articles from the Taylor & Francis database.

TABLE 3. Inclusion and exclusion criteria.

Criteria	Inclusion	Exclusion
Period	Publication year within the timeframe of 2021 to 2024.	Publication year before 2021 and After 2024
Type of publication	Journal	Conference, Proceedings, E-book, Report, and other.
Language	Written language in English	Written in other languages than English
Accessibility	Full-text paper	Could not have access to full-text paper
Type of Study	Technical Paper or Research Paper	Review Paper
Relevance to Topic	Focus and relevant security in multi-cloud or multi-cloud security	Does not concentrate and is irrelevant to multi-cloud security or multi-cloud security.

**FIGURE 3.** Flowchart for study selection.

A total of 2038 articles at this identification phase. The Parsifal tool was employed for subsequent screening of duplicate studies to automatically detect and eliminate redundant articles. The screening process was conducted by implementing all the inclusion criteria. Subsequently, proceed to the second phase, which involves screening by evaluating the title and abstract to determine the article's relevance, along with additional factors such as methodology and empirical experiments.

Then in the third stage, namely eligibility to ensure the quality of the articles that pass the screening are of good or high quality, by assessing 53 articles based on the answers to 10 quality assessment questions as stated in the Quality Assessment sub section. if the article gets a final score of more than 7 then it will be considered passed. So the final

result of this stage is 30 articles that have a score of more than 7 enter the next stage, namely included in the data extraction process.

D. QUALITY ASSESSMENTS

A fundamental component of any systematic literature review (SLR) is the evaluation of quality. This evaluation entails assessing the principal research incorporated in the review strategy. The objective of quality assessment is to guarantee the precision and dependability of the SLR findings. Quality evaluation can be conducted by administering a survey comprising a series of questions. This aligns with the recommendation [7]:

- 1) Does this study address specific security solutions for multi-cloud?
- 2) Does this study present empirical data or experimental results?
- 3) Does this study describe the implementation of the security solution in detail?
- 4) Is the research methodology appropriate for testing the effectiveness of a security solution?
- 5) Does this study include performance analysis or evaluation of security solutions?
- 6) Does this study present a clear security architecture?
- 7) Did this study use validation techniques such as simulation or field testing?
- 8) Does this study provide information about limitations and potential improvements?
- 9) Does this study have conclusions supported by data?
- 10) Does this study provide technical recommendations for implementing multi-cloud security?

TABLE 4. Cut-Off quality assessment.

Category	Description	Score
Yes	The information is presented clearly and explicitly.	1.0
Partially	The information is presented clear but implicit manner.	0.5
No	Information cannot be inferred.	0.0

The answers to the quality assessment questions will be categorized into three, each of which has a description and score as shown in Table 4. The question will receive a score of 1 if the answer is clear and explicit stating that there is a comprehensive and understanding explanation. While the question will receive a score of 0.5 if it receives a partial or incomplete answer and is clear but implicit. In addition, if the question does not receive a answer, it will get a score of 0. This highlights the importance of articles determined by the response to the question. The maximum score is 10 according to the total of 10 questions used in the quality assessment stage, while the minimum score is 0. Each article at this stage will be studied carefully and thoroughly and then it will be decided which articles meet the threshold value of more than 7 will be declared to have passed this stage. A threshold

value of more than 7 is used to ensure the quality of articles through a strict procedure by requiring all information to meet a number of requirements, which shows dedication to superior standards. Based on a total of 53 articles reviewed at this stage, 23 articles were excluded because they did not meet the threshold value, so only 30 articles were selected for further careful reading and analysis.

E. DATA EXTRACTION

The compiled data extraction form was used during the Kitchenham Guide, which can be seen in Table 5. To help find the position of relevant information in a primary study document, the Humata tool was used. The Highlight Citations feature in Humata allows one to search for the position of information in the document section that contains information according to the keywords submitted, thereby speeding up the content search process.

However, it is important to emphasize that Humata is not used to automatically extract the data. Instead, the data extraction process is still carried out manually by reading and carefully reviewing the context of the highlighted section by Humata and then evaluating and recording the appropriate information in the data extraction form. This approach aims to maintain accuracy, reduce potential bias from using the Humata tool, and ensure that all entered data have undergone a manual verification process.

Table 5 is a guide used to extract data from the various primary studies. This table divides the data into two main types: qualitative data and contextual data. Qualitative data includes descriptive information, such as the title of the study, authors, year of publication, and a general summary. Meanwhile, contextual data provides a more in-depth picture of the study, including the problem area studies, solutions offered, challenges faced, and recommendations for future research directions.

TABLE 5. Data extracted from each primary study.

Data Type	Data Item	Description
Qualitative data	Title	Title of the primary study
	author	Author of the study
	Year published	Publication year of study
	Article rank	i.e. Q1, Q2, Q3, Q4
Contextual data	Summary	Summary
	Issue area	Cryptography, data, and other
	Solution	Currently solution for issue solving
	Challenges	Challenges in research
	Future direction	Recommendation research

Using this table, can systematically collect and organize data form various sources, making it easier to analyze and compare between studies. In simple terms, this table serves as a framework that ensures that all important information from each study is recorded properly and consistently. The data is then used to answer research questions. The data is presented in a table format, making it easier to analyze further.

F. DATA SYNTHESIS

Data synthesis seeks to amalgamate information from chosen studies to address the research question presented. Occasionally, corroborative evidence from various sources can bolster an argument, even when an individual study lacks adequate substantiation. This synthesis process entails gathering data in both quantitative and qualitative formats, subsequently employing a narrative methodology to structure the data in alignment with the research objectives. The findings are displayed in tables aligned with the research question, or other visualization methods, including bar graphs, pie charts, and tables, to elucidate multi-cloud security issues, existing solutions, challenges, and research directions.

III. RESULT AND DISCUSSION

This section presents the findings and includes a full explanation. The document commences with an overview of the distribution of research papers, subsequently addressing the findings pertinent to each research question.

A. RESULT OF STUDIES

This review synthesized scientific journals published from 2021 to 2024. A total of 30 journals were acquired from the preceding stage. Figure 4 illustrates the distribution of these journals according to their year of publication. Ten journals were published in 2021, four in 2022, nine in 2023, and seven in 2024. The findings demonstrate a sustained interest in the domain of multi-cloud security.

B. DISCUSSION OF STUDIES

The current trends in security challenges within multi-cloud computing are listed in Table 6, where the taxonomy area was adopted from Kushala and Shylaja [3]. Here, we combined data storage and data sanitization into one area with the name data. Thus there are now six areas, namely discussing issues related to Data, Cryptography, Virtualization, Confidential Computing, Authentication and Authorization, and Service Availability. Table 6 below consolidates various related studies, which we have classified according to six taxonomies of

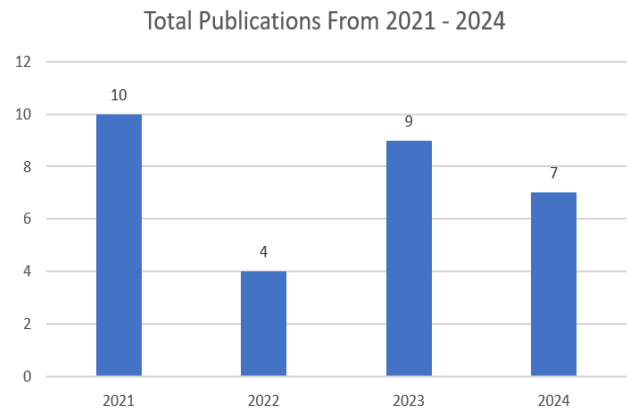


FIGURE 4. Total publications from 2021 to 2024 (Final articles).

TABLE 6. Trend security issue in multi-cloud.

No	Name Author	Year	Area					
			Data	Cryptography	Virtualization	Confidential Computing	Authentication and Authorization	Service Availability
1	Hang Zhang et al [8]	2024		✓		✓		
2	Sumeet Kaur et al [9]	2024	✓	✓			✓	
3	Manjanyaik et al [10]	2024	✓	✓			✓	
4	Ze Yang et al [11]	2024		✓		✓	✓	
5	Simarjeet Makkar et al [12]	2024						✓
6	M. A. Altahat et al [13]	2024		✓				
7	Wenchao Li et al [14]	2024	✓	✓				
8	Yakoob et al [15]	2023	✓	✓			✓	
9	Xiaolu Zhang et al [16]	2023	✓				✓	
10	Mahender Kumar et al [17]	2023	✓	✓		✓	✓	
11	Wenjuan Li et al [18]	2023						✓
12	Rudolf Lovrencic and Dejan Škvorc [19]	2023	✓			✓		✓
13	Guangcan Yang et al [20]	2023	✓	✓			✓	
14	Mingxing zhou et al [21]	2023		✓			✓	
15	K. Jyothsna Devi et al [22]	2023	✓	✓				
16	Jiawei Zhang et al [23]	2023					✓	
17	Cheng Zhang et al [24]	2022	✓	✓			✓	✓
18	Jiguo Li et al [25]	2022	✓	✓			✓	✓
19	Qing Wu et al [26]	2022	✓	✓			✓	
20	Qi Li et al [27]	2022	✓			✓	✓	
21	Chenlin Huang et al [28]	2021		✓			✓	✓
22	Sunitha Pachala et al [29]	2021	✓	✓				✓
23	Fatma Lahmar et al [30]	2021	✓	✓			✓	
24	Hussain et al [31]	2021				✓	✓	✓
25	Ahad Niknia et al [32]	2021	✓	✓				✓
26	K.A. Torkura et al [33]	2021	✓				✓	
27	N. R. Rejin Paul and D. Paul Raj [34]	2021					✓	
28	Reda Maher and Omar A. Nasr [35]	2021	✓					
29	Matthew Dickinson et al [36]	2021				✓		
30	S. D. C. di Vimercati et al [37]	2021	✓					

security issue areas, to facilitate a comprehensive understanding of security concerns in multi-cloud in each area.

Table 6 delineates the annual research emphasis on security within multi-cloud environments by topic. Each row denotes a work that tackles one or more security concerns, including data security, encryption, virtualization, confidential computing, authentication and authorization, and service availability. By marking (✓) a specific column, we can swiftly discern the primary domains of research interest. This chart presents

a comprehensive overview of the range and complexity of security issues addressed, highlighting evolving patterns from an initial emphasis on data protection and cryptography to emerging concerns such as secret computing.

Data security, encompassing cryptography, together with authentication and authorization, are the most commonly addressed topics in numerous research, as indicated by the frequency of checkmarks (✓) in the column. Nevertheless, upon deeper examination, the domains of virtualization,

TABLE 7. Summary of problems, solutions, challenges, and recommendations.

No	Name Author	Year	Summary			
			Problems	Solutions	Challenges	Recommendations
1	Hang Zhang et al [8]	2024	Data Privacy Risks, difficulty of resource allocation efficiency, reliance on a single serverless architecture	Serverless multi-cloud edge computing security model which uses a combination AES & RSA Action-Constrained DRL	DRL model scalability if managing large data and resource, ensure interoperability between cloud with different attributes and standard	Improved interoperability, and Improved DRL Techniques
2	Sumeet Ka ur et al [9]	2024	Vulnerability in server-to server, inefficient key management, insecure authentication issues	Attribute-Based Encryption with Outsourced Decryption (ABE-OD), SIDH Protocol, Blowfish and RSA, BLAKE2, proxy server model and trusted authorities	Ability to adapt to new threats, security algorithm optimization	Integration of AI-based approach, model expansion to real-world scenarios encryption algorithm optimization
3	Manjyanaik et al [10]	2024	Vulnerable data confidentiality, limitations of traditional encryption algorithms, difficult to track data stored across multiple CSPs	Improved RSA technique, using AWS S3 with support client side encryption, dan AWS IAM, splitting files into small fragments	Encryption algorithm efficiency, complex access management, resistance to quantum threats, security model scalability	Improved Rivest Shamir Adleman (IRSA) optimization, Development of Quantum Resistant Cryptographic Algorithm, AI integration
4	Ze Yang et al [11]	2024	Limitations of traditional authentication, efficiency issues, wireless communication vulnerabilities	Lattice-based authentication scheme, QKD, Quantum security service cloud	Resistance to quantum computing, heterogeneous network integration, real world implementation	Use of quantum resistant blockchain, fog computing integration, stability test development
5	Simarjeet Makkar et al [12]	2024	Overloading and possible service failures, risk variations and management complexities, cost and efficiency versus security, absence of centralized and real-time monitoring	Scheduling and resource allocation optimum with HE-CGA, Security-SLA model, considering time & cost metrics for job scheduling, IAM integration and access policies	Scalability and CSP heterogeneity, aligning QoS objectives with security, dynamic security-SLA measurement, automated threat detection & response, lack of details regarding practical implementation	Job scheduling integration with security monitoring, adaptive & context-aware SLA, advanced optimization using AI, real-world implementation customization
6	M. A. Altahat et al [13]	2024	Data security during migration, variation of SLA requirements & encryption needs, complex scheduling & balancing, ensuring confidentiality in heterogeneous environments	Hybrid encryption, two stage container scheduling, multi-objective optimization, ranking-based encryption scheme selection	Incorporating cryptographic overhead into scheduling models, scalability in large-scale data center environments, hardware heterogeneity & security SLAs, managing migration load across data center	formulating ML-based scheduling approach, incorporating erasure coding, and emphasizing the protection of in-use and at-rest data to enhance the security and efficiency of the multi-cloud ecosystem
7	Wenchao Li et al [14]	2024	Data integrity and storage reliability, data confidentiality against cloud and malicious users, copy consistency testing without downloading & decrypting,	Use a Verified Public Key Encryption with Equality Test (PKEET), limitation of test rights, homomorphic extension support, public verifiability & proof generation	Encryption scheme facilitating equality testing without revealing plaintext, public verification with computational efficiency, ensuring security against diverse attack models	Strengthening IND-CCA Security in PKEET, dynamic operations and real-world scalability support, and collaborating on standardization and industrial implementation
8	Yakoob et al [15]	2023	Threats to sensitive data, lack of data confidentiality, byzantine server failure	Byzantine protocol, shamir secret sharing method, Depsky architecture, and the third combination	Operational efficiency, interoperability across cloud, complex key management	Develop more efficient encryption algorithm, identity-based access protocol enhancements
9	Xiaolu Zhang et al [16]	2023	Vulnerabilities, Unauthorized access, and difficulty detecting threats in multi-cloud	Process mining approach with full event log, conformance checking, and multi-dimensional process analysis, use ML in IDS, also use attribute-based security	Complexity of distributed log collection, coordination across cloud service providers (CSPs), internal threat detection, full lifecycle analysis	Develop more precise analysis tools, AI automate detection enhancements
10	Mahender Kumar et al [17]	2023	Replacement attacks, lack of transparency from cloud service providers (CSPs), data confidentiality breach	Identity-based integrity auditing protocol, identity-based blind signature, use of third-party auditors, batch verification.	Replacement attack migration, scalability limitation, trust in CSPs	Development standard framework for multi-auditing for interoperability and trust between different CSPs
11	Wenjuan Li et al [18]	2023	Lack of cloud interface standards, dependence on particular service	Distributed broker model and two layer Restricted Monte Carlo Tree Search	real-time service selection optimization, interoperability across cloud providers, cloud	Cloud federation model expansion, wider cloud API integration

TABLE 7. (Continued.) Summary of problems, solutions, challenges, and recommendations.

			provider, difficulty in service selection	(RMCTS) for learn user preferences and offer recommendation, cloud federation platform	federation reliability and security	development, RMCTS algorithm efficiency improvements
12	Rudolf Lovrencic and Dejan Škvorc [19]	2023	Trust in CSPs, compromise of data privacy during storage and processing	Data and code fragmentation, multi-cloud computation pattern, constraint-based security modelling	Computational and communication overhead, improve fragmentation efficiency, dynamic security constraint management	Automated fragmentation process, integration with homomorphic encryption, and security aspects
13	Guangcan Yang et al [20]	2023	Data leaks due to attacks on a single CSP, decryption computational burden, reliance on a single trusted entity, lack of data security classification	Attribute-Based Encryption with Data Security Classification (ABE-DSC) with data storage across multiple CSP, CP-ABE and outsourcing description	Computational efficiency, complex key management, resistance to attribute-based plaintext attacks, interoperability between CSPs	Improve efficiency of outsourced decryption algorithm, AI-Based key management automation, integration with block-chain, development more flexible multi-level security model
14	Mingxing zhou et al [21]	2023	Deficiency of trust among cloud units, vulnerabilities in virtualization & cloud platform, absence of integrity verification mechanism, performance burden	Trusted computing based on TPM/vTPM, D-dual level verification, cryptography approach (SM2, SM3, SM4), optimized workflow for remote attestation	Balancing security level with operational overhead, 6G architecture scalability and complexity, security at the virtualization layer, trust coordination across cloud units & providers	optimization and parallelization for remote attestation, further study on performance & security trade-off of SMx algorithm, integration with zero-trust architecture
15	K. Jyothsna Devi et al [22]	2023	Data vulnerabilities in multi-cloud, lack of data confidentiality and integrity, attacks on shares schemas, need for low latency and computation	Data hiding/reversible watermarking, double scan pixel position shuffling, neighbor mean interpolation for embedding	Balancing security with computational overhead, resistance to advanced attacks (geometrical attacks) shares and key management schemes,	Improving resistance to geometrical attacks, using DL for embedding optimization & attack detection, a more secure & distributed key management structure
16	Jiawei Zhang et al [23]	2023	Lack of a cohesive framework for SSO and collaboration, inadequate user access management in multi-cloud environment, difficulty in evaluating quality and trustworthiness of service	Role-based trust assessment, efficient multi-cloud single sign-on, multi-cloud collaborative framework, implementation and performance analysis	Precise service trust modeling, secure and lightweight SSO authentication, cross-domain collaboration with autonomy and security, bidirectional access patch integrity	Integration of ML for the detection of malicious users/providers, extension of access control to attribute-based encryption, standardization of SSO token mechanism and support for dynamic configuration
17	Cheng Zhang et al [24]	2022	Loss of data integrity, reliance on third parties, difficulty in arbitrating disputes	Blockchain usage, batch data audit, smart contracts	Identifying rogue CSPs, limitation of blockchain	Development of backup audit mechanism, improved blockchain technology or techniques
18	Jiguo Li et al [25]	2022	Loss of direct control over data, data integrity vulnerabilities, collusion between cloud servers	Identity-based provable data possession (PDP) scheme, distributing data copies across multiple cloud servers	Computation and communication efficiency, collaboration between cloud servers, schema scalability	Developing schemas that support dynamic data, verification algorithm optimization, AI-based distributed data management
19	Qing Wu et al [26]	2022	Single authority vulnerability, cloud server reliability, access privacy limitations	Uses of consortium blockchain, Multi-authority authority, access and search privacy	Complexity of attribute management, Computational overhead on blockchain, search efficiency	Multi-keyword search development, Integration of additional features on the consortium blockchain
20	Qi Li et al [27]	2022	Lack of data accountability, Vulnerability to centralized authority, log insecurity	Cloud-blockchain fusion framework, Hyperledger fabric blockchain usage, secure log design	Interoperability across cloud providers, efficiency of logging and tracking mechanism	Blockchain efficiency improvement, AI technology integration
21	Chenlin Huang et al [28]	2021	Loss of user control over the computing environment, policy inconsistencies during VM migration	Policy customized trusted cloud service (PC-TCS) architecture, customized vm migration protocol	VM migration efficiency, data security during migration, interoperability across cloud providers	ABS and blockchain performance optimization, integration with container services like docker
22	Sunitha Pachala et al [29]	2021	Insider attacks, dependence on a single CSP	Hybrid approach like a depsky architecture, proxy re-encryption	Data management complexity, scalability	Implementation of stronger encryption method, AI-based approach
23	Fatma Lahmar et al [30]	2021	Inter-cloud mistrust, lack of service composition security, difficulty identifying secure services	Use of fuzzy formal concept analysis, rough set theory, secure multi-cloud service composition	Security management complexity, efficiency of service composition algorithm	Optimization of fuzzy FCA dan rough set algorithm, development of universal security standard

TABLE 7. (Continued.) Summary of problems, solutions, challenges, and recommendations.

24	Hussain et al [31]	2021	User metadata risk, vulnerabilities in password-based authentication	Implementing multi-factor authentication (MFA), single sign on usage, AAAA Security model	Cross vendor security solution integration, user metadata privacy	MFA and SSO Performance optimizations, biometric-based authentication enhancements.
25	Ahad Niknia et al [32]	2021	Dependence on a single provider, insecurity of traditional block encryption, space efficiency of traditional sharing schemes	Practical and Efficient Schemes for Secret Sharing (PRESS), SCUD storage service, use of ECDSA digital signature	Computational complexity of polynomial, optimal parameter management	Development of more efficient interpolation, dynamic storage ration adjustment, exploring the use of ai storage management
26	K.A. Torkura et al [33]	2021	Misconfiguration Vulnerabilities, Human errors & cloud-specific attacks	Proposed system: CSBAuditor, Security-focused configuration management (SecCM)	Lack of reference systems & data for CSPM, reliance on APIs & Non-real-time logs	Cloud scope & service expansion, further incident response and logging/monitoring integration, CSPM dataset standardization and availability
27	N. R. Rejin Paul and D. Paul Raj [34]	2021	Lack of comprehensive trust mechanism, unauthorized access risk, potential data leakage and data integrity, limitations of the single-cloud model in a multi-cloud environment	Trust-based access control on user & server, cyclic shift transposition algorithm for data encryption, server selection based on trust degree, performance evaluation & experiments	Testing and validating dynamic trust models, balancing computational overhead with security, integration across cloud providers, dealing with advanced attacks	Enhance trust model development, integrate adaptive encryption & trust-based access control, standardize cross-cloud, optimize overhead and scalability, develop anti-collusion mechanism
28	Reda Maher and Omar A. Nasr [35]	2021	Privacy and data security, single-provider lock-in, multi-cloud management complexity, reliability & disaster recovery	Encryption & data partitioning (data chunking), replication & incremental versioning, fog computing “Droplet” as abstraction layer, key & metadata management in fog node	Heterogeneity of APIs & cloud services, balance between replication versus cost and security, resource & bandwidth management, effective disaster recovery, edge-fog integration & scalability	Improved scheduling & QoS-Aware backup, use of linear block codes, integration with real-time threat detection, further optimization & evaluation in real conditions
29	Matthew Dickinson et al [36]	2021	Absence of formal method to address security requirements across domains, conflict between performance and security requirement, diversity of security policies among cloud environments	NIST-Based security requirements formalization (SSpecs), Policy alignment mechanism (Portunes algebra), joint performance-security optimization model, implementation in OnTimeURB	Comprehensive security formalization, performance security trade-off, complexity of heterogeneous domain policy management (RSpecs), scalability of resource allocation, risk and reliability evaluation	Automated end-to-end security integration in multi-cloud DevOps, Utilizing ML for adaptive scheduling, expanding NIST model for addressing emerging threat, integration trusted environments, and standardization of security policies.
30	S. D. C. di Vimercati et al [37]	2021	Variations in security levels and features of each cloud service, difficulty in determining security suitability for data granularity, ensuring confidentiality in multi-service schemes	Formal model of security properties, owner-side encryption, resource-based granular protection, global allocation requirements, problem modeling with binary programming	Integrating diverse security requirements, developing an allocation model that satisfies multiple constraints, compatibility considerations and additional services	Expansion to broader security attributes, scalable heuristic & approximation, management of runtime adaption and dynamic requirements, integration with enhanced multi-cloud orchestration

confidential computing and service availability remain infrequently addressed, indicating substantial research potential in these areas.

A number of papers in the table serve as exemplars illustrating the diversity of methodologies and subjects addressed. For instance, The study [24] utilized blockchain technology to ensure the security and efficient auditing of data stored in the cloud. Blockchain technology ensures data integrity through a transparent and immutable auditing system. While [18] noted that users encounter challenges in selecting optimal cloud services due to the multitude of options and the dynamic nature of the cloud environment. In addition, the study [36] proposes a formal model based on

NIST and portunes algebra mechanism to normalize policies across different domains, also align security and performance requirements in multi-cloud environments. Thus, Table 6 not only summarizes the security issues that exist in multi-cloud, but also show a map of research developments that are moving and cloud differ from 2021-2024.

Table 7 summarizes the issues faced in multi-cloud security, the solutions implemented, the challenges of applying those solutions, and recommendations for further research based on recent studies such as those by Zhang et al. [8], Kaur et al. [9], Manjyanik et al [10], Yang et al. [11], and others. Overall, this table helps to understand the patterns and trends of security solutions and to asses the research

gaps that are still open. Many researchers emphasize the need to improve security technologies including quantum-resistant encryption, blockchain, and fog computing.

Moreover, factors of complexity and adaptability are significant. Numerous studies emphasize the necessity of equilibrating security and efficiency to prevent the implemented solution from overburdening the system. Blockchain-based solutions offer enhanced transparency and traceability; however, they incur overheads regarding scalability and performance. Consequently, numerous recommendations advocate for the implementation of hybrid methodologies, such as integrating blockchain infrastructure with alternative systems (e.g., IPFS), or employing quantum-resistant blockchain to address prospective threats.

Conversely, data protection at scale across multiple providers (multi-cloud) constitutes a distinct issue. Numerous studies indicate that reliance on a sole provider (CSP) heightens the risk of data breaches and diminishes the control of data proprietors. Consequently, various solutions advocate for the utilization of attribute-based encryption schemes (CP-ABE) in conjunction with a multi-cloud model, alongside the execution of distributed security policies supported by dynamically managed container services and virtual machine.

The notion of AI-driven methodologies, including automated conformance verification, AI-enhanced key management, and anomaly identification, is gaining prominence as a means to enhance threat detection and immediate security response. Integrating artificial intelligence technology enables intelligent and adaptive execution of key management and verification mechanisms. The deployment of AI is anticipated to simplify the encryption process, identify insider threats, and enhance the optimization of security resource capacity in response to evolving attack patterns.

Thus, table 7 illustrates the complex dynamics present in the realm of multi-cloud security by highlighting various problems, solutions, challenges, and recommendations. Issues such as data privacy, integrity, accountability key management, service selection, misconfiguration, various vulnerability and threat have been addressed with the emergence of various solutions like development various encryption algorithms, allocation model, trust security model, machine learning, and other solutions. However, researchers still face various challenges such as interoperability, scalability, and standardization. Future developments recommendations are diverse, such as the use of AI/DL, and real-world scenarios so that it can be widely implemented by various organizations adopting multi-cloud strategies.

IV. CONCLUSION

Based on 30 reviewed studies, it can be seen that the trend of multi-cloud security issues from 2021 to 2024 is mostly directed at the areas of data, cryptography, authentication and authorization. Meanwhile, the areas of virtualization, confidential computing, and service availability still have aspects that can be explored further for future research. Researchers are exploring various solutions to address security issues

in multi-cloud, including data vulnerability, encryption deficiencies, and inadequate authentication-authorization mechanisms. These solutions include blockchain, quantum-resistant cryptography methods, and AI-driven strategies. However, efficiency and scalability are challenges, as advanced security mechanisms require significant computing resources.

Future progress requires more comprehensive security standards to facilitate interoperability and collaboration among stakeholders, especially in the application of innovative technologies such as lightweight cryptography, AI threat intelligence, and decentralized verification. This underscores the need to advance and improve security technologies in a multi-cloud ecosystem, which is in line with the increasing use of cloud computing services in various sectors. There are also a number of limitations in this systematic literature review research, such as the scope of publication in only 30 primary studies in a certain period, so there may be other studies with new approaches that may be missed. Second, this study does not explicitly discuss real-world deployment or industrial case studies that provide a practical perspective, so that recommendations for further systematic literature review research can complement the limitations of this study.

REFERENCES

- [1] M. R. M. Assis and L. F. Bittencourt, "A survey on cloud federation architectures: Identifying functional and non-functional properties," *J. Netw. Comput. Appl.*, vol. 72, pp. 51–71, Sep. 2016, doi: [10.1016/j.jnca.2016.06.014](https://doi.org/10.1016/j.jnca.2016.06.014).
- [2] P. R. Shukla and V. M. Patil, "A comprehensive review of frameworks for achieving interoperability in multi-cloud environments," in *Proc. 2nd Int. Conf. Informat. (ICI)*, Nov. 2023, pp. 1–6, doi: [10.1109/ici60088.2023.10421703](https://doi.org/10.1109/ici60088.2023.10421703).
- [3] M. V. Kushala and B. S. Shylaja, "Recent trends on security issues in multi-cloud computing: A survey," in *Proc. Int. Conf. Smart Electron. Commun. (ICOSEC)*, India: IEEE, Sep. 2020, pp. 777–781, doi: [10.1109/ICOSEC49089.2020.9215303](https://doi.org/10.1109/ICOSEC49089.2020.9215303).
- [4] R. Kaur, D. Anand, U. Kaur, and S. Verma, "Analysis and evaluation of bio-inspired algorithmic framework, potential application in cloud/multi-cloud environment," in *Proc. IEEE 5th Int. Conf. Cybern., Cognition Mach. Learn. Appl. (ICCCMLA)*, Oct. 2023, pp. 354–361, doi: [10.1109/ICCCMLA58983.2023.10346828](https://doi.org/10.1109/ICCCMLA58983.2023.10346828).
- [5] A. Alajroush, R. Latif, and T. Saba, "Trust management frameworks in multi-cloud environment: A review," in *Proc. 6th Int. Conf. Women Data Sci. at Prince Sultan Univ. (WiDS PSU)*, Mar. 2023, pp. 117–120, doi: [10.1109/WiDS-PSU57071.2023.00034](https://doi.org/10.1109/WiDS-PSU57071.2023.00034).
- [6] M. Allakonda and K. Sagar, "A survey on data security challenges in multi cloud environment," in *Proc. IEEE Int. Conf. Electron., Comput. Commun. Technol. (CONECCT)*, Jul. 2021, pp. 1–5, doi: [10.1109/CONECCT52877.2021.9622722](https://doi.org/10.1109/CONECCT52877.2021.9622722).
- [7] B. Kitchenham and S. M. Charters, (2007). *Guidelines for Performing Systematic Literature Reviews in Software Engineering*. [Online]. Available: <https://www.researchgate.net/publication/302924724>
- [8] H. Zhang, J. Wang, H. Zhang, and C. Bu, "Security computing resource allocation based on deep reinforcement learning in serverless multi-cloud edge computing," *Future Gener. Comput. Syst.*, vol. 151, pp. 152–161, Feb. 2024, doi: [10.1016/j.future.2023.09.016](https://doi.org/10.1016/j.future.2023.09.016).
- [9] S. Kaur, M. D. Khare, M. Bhatt, V. Haripriya, A. Kumar, and A. Singla, "Multi-cloud security model: Establishment of inter-server communication for authentication integrity," *Int. J. Syst. Assurance Eng. Manage.*, Oct. 2024, doi: [10.1007/s13198-024-02551-0](https://doi.org/10.1007/s13198-024-02551-0).
- [10] H. N. B. Manjyanaik, R. Mohanty, and J. M. Kannan, "Preserving confidential data using improved Rivest-Shamir Adleman to secure multi-cloud," *Int. J. Intell. Eng. Syst.*, vol. 17, no. 4, pp. 162–171, 2024, doi: [10.22266/IJIES2024.0831.13](https://doi.org/10.22266/IJIES2024.0831.13).

- [11] Z. Yang, Q. Shi, T. Cheng, X. Wang, R. Zhang, and L. Yu, "A security-enhanced authentication scheme for quantum-key-distribution (QKD) enabled Internet of Vehicles in multi-cloud environment," *Veh. Commun.*, vol. 48, Aug. 2024, Art. no. 100789, doi: [10.1016/j.vehcom.2024.100789](https://doi.org/10.1016/j.vehcom.2024.100789).
- [12] S. Makkar, J. Sidhu, T. Zaidi, R. Batra, P. Garg, and J. Shekhawat, "Advanced model for maximizing multi-cloud security through job scheduling," *Int. J. Syst. Assurance Eng. Manage.*, vol. 2024, pp. 1–9, Nov. 2024, doi: [10.1007/s13198-024-02567-6](https://doi.org/10.1007/s13198-024-02567-6).
- [13] M. A. Altahtat, T. Daradkeh, and A. Agarwal, "Optimized encryption-integrated strategy for containers scheduling and secure migration in multi-cloud data centers," *IEEE Access*, vol. 12, pp. 51330–51345, 2024, doi: [10.1109/ACCESS.2024.3386169](https://doi.org/10.1109/ACCESS.2024.3386169).
- [14] W. Li, W. Susilo, C. Xia, L. Huang, F. Guo, and T. Wang, "Secure data integrity check based on verified public key encryption with equality test for multi-cloud storage," *IEEE Trans. Dependable Secure Comput.*, vol. 21, no. 6, pp. 5359–5373, Nov. 2024, doi: [10.1109/TDSC.2024.3375369](https://doi.org/10.1109/TDSC.2024.3375369).
- [15] S. K. Yakob and V. K. Reddy, "Efficient identity-based multi-cloud security access control in distributed environments," *Int. J. e-Collaboration*, vol. 19, no. 3, pp. 1–13, Jan. 2023, doi: [10.4018/ijec.316771](https://doi.org/10.4018/ijec.316771).
- [16] X. Zhang, L. Cui, W. Shen, J. Zeng, L. Du, H. He, and L. Cheng, "File processing security detection in multi-cloud environments: A process mining approach," *J. Cloud Comput.*, vol. 12, no. 1, pp. 1–18, Jul. 2023, doi: [10.1186/s13677-023-00474-y](https://doi.org/10.1186/s13677-023-00474-y).
- [17] M. Kumar, C. Maple, and S. Chand, "An efficient and secure identity-based integrity auditing scheme for sensitive data with anti-replacement attack on multi-cloud storage," *J. King Saud Univ. - Comput. Inf. Sci.*, vol. 35, no. 9, Oct. 2023, Art. no. 101745, doi: [10.1016/j.jksuci.2023.101745](https://doi.org/10.1016/j.jksuci.2023.101745).
- [18] W. Li, J. Cao, B. Zhou, S. Deng, Q. Zhang, K. Hu, J. Li, and H. Zhao, "Multi-cloud service provision based on decision tree and two-layer restricted Monte Carlo tree search," *Internet Things*, vol. 22, Jul. 2023, Art. no. 100751, doi: [10.1016/j.iot.2023.100751](https://doi.org/10.1016/j.iot.2023.100751).
- [19] R. Lovrenčić and D. Škvorc, "Multi-cloud applications: Data and code fragmentation for improved security," *Int. J. Inf. Secur.*, vol. 22, no. 3, pp. 713–721, Jun. 2023, doi: [10.1007/s10207-022-00658-8](https://doi.org/10.1007/s10207-022-00658-8).
- [20] G. Yang, P. Li, K. Xiao, Y. He, G. Xu, C. Wang, and X. Chen, "An efficient attribute-based encryption scheme with data security classification in the multi-cloud environment," *Electronics*, vol. 12, no. 20, p. 4237, Oct. 2023, doi: [10.3390/electronics12204237](https://doi.org/10.3390/electronics12204237).
- [21] M. Zhou, P. Xiao, Q. Wang, S. Ruan, X. Chen, and M. Yang, "Enhancing the trustworthiness of 6G based on trusted multi-cloud infrastructure: A practice of cryptography approach," *Comput. Model. Eng. Sci.*, vol. 138, no. 1, pp. 957–979, 2024, doi: [10.32604/cmes.2023.028612](https://doi.org/10.32604/cmes.2023.028612).
- [22] K. J. Devi, P. Singh, J. K. Dash, H. K. Thakkar, S. Tanwar, and A. Alabdulatif, "Secure transmission of medical images in multi-cloud e-healthcare applications using data hiding scheme," *J. Inf. Secur. Appl.*, vol. 79, Dec. 2023, Art. no. 103655, doi: [10.1016/j.jisa.2023.103655](https://doi.org/10.1016/j.jisa.2023.103655).
- [23] J. Zhang, T. Li, Z. Ying, and J. Ma, "Trust-based secure multi-cloud collaboration framework in cloud-fog-assisted IoT," *IEEE Trans. Cloud Comput.*, vol. 11, no. 2, pp. 1546–1561, Apr. 2023, doi: [10.1109/TCC.2022.3147226](https://doi.org/10.1109/TCC.2022.3147226).
- [24] C. Zhang, Y. Xu, Y. Hu, J. Wu, J. Ren, and Y. Zhang, "A blockchain-based multi-cloud storage data auditing scheme to locate faults," *IEEE Trans. Cloud Comput.*, vol. 10, no. 4, pp. 2252–2263, Oct. 2022, doi: [10.1109/TCC.2021.3057771](https://doi.org/10.1109/TCC.2021.3057771).
- [25] J. Li, H. Yan, and Y. Zhang, "Efficient identity-based provable multi-copy data possession in multi-cloud storage," *IEEE Trans. Cloud Comput.*, vol. 10, no. 1, pp. 356–365, Jan. 2022, doi: [10.1109/TCC.2019.2929045](https://doi.org/10.1109/TCC.2019.2929045).
- [26] Q. Wu, T. Lai, L. Zhang, Y. Mu, and F. Rezaeiabagha, "Blockchain-enabled multi-authorization and multi-cloud attribute-based keyword search over encrypted data in the cloud," *J. Syst. Archit.*, vol. 129, Aug. 2022, Art. no. 102569, doi: [10.1016/j.sysarc.2022.102569](https://doi.org/10.1016/j.sysarc.2022.102569).
- [27] Q. Li, Z. Yang, X. Qin, D. Tao, H. Pan, and Y. Huang, "CBFF: A cloud-blockchain fusion framework ensuring data accountability for multi-cloud environments," *J. Syst. Archit.*, vol. 124, Mar. 2022, Art. no. 102436, doi: [10.1016/j.sysarc.2022.102436](https://doi.org/10.1016/j.sysarc.2022.102436).
- [28] C. Huang, W. Chen, L. Yuan, Y. Ding, S. Jian, Y. Tan, H. Chen, and D. Chen, "Toward security as a service: A trusted cloud service architecture with policy customization," *J. Parallel Distrib. Comput.*, vol. 149, pp. 76–88, Mar. 2021, doi: [10.1016/j.jpdc.2020.11.002](https://doi.org/10.1016/j.jpdc.2020.11.002).
- [29] S. Pachala, C. Rupa, and L. Sumalatha, "An improved security and privacy management system for data in multi-cloud environments using a hybrid approach," *Evol. Intell.*, vol. 14, no. 2, pp. 1117–1133, Jun. 2021, doi: [10.1007/s12065-020-00555-w](https://doi.org/10.1007/s12065-020-00555-w).
- [30] F. Lahmar and H. Mezni, "Security-aware multi-cloud service composition by exploiting rough sets and fuzzy FCA," *Soft Comput.*, vol. 25, no. 7, pp. 5173–5197, Apr. 2021, doi: [10.1007/s00500-020-05519-x](https://doi.org/10.1007/s00500-020-05519-x).
- [31] M. I. Hussain, J. He, N. Zhu, F. Sabah, Z. A. Zardari, S. Hussain, and F. Razque, "AAAA: SSO and MFA implementation in multi-cloud to mitigate rising threats and concerns related to user metadata," *Appl. Sci.*, vol. 11, no. 7, p. 3012, Mar. 2021, doi: [10.3390/app11073012](https://doi.org/10.3390/app11073012).
- [32] A. Niknia, M. Correia, and J. Karimpour, "Secure cloud-of-clouds storage with space-efficient secret sharing," *J. Inf. Secur. Appl.*, vol. 59, Jun. 2021, Art. no. 102826, doi: [10.1016/j.jisa.2021.102826](https://doi.org/10.1016/j.jisa.2021.102826).
- [33] K. A. Torkura, M. I. H. Sukmana, F. Cheng, and C. Meinel, "Continuous auditing and threat detection in multi-cloud infrastructure," *Comput. Secur.*, vol. 102, Mar. 2021, Art. no. 102124, doi: [10.1016/j.cose.2020.102124](https://doi.org/10.1016/j.cose.2020.102124).
- [34] N. R. Rejin Paul and D. Paul Raj, "Enhanced trust based access control for multi-cloud environment," *Comput., Mater. Continua*, vol. 69, no. 3, pp. 3079–3093, 2021, doi: [10.32604/cmc.2021.018993](https://doi.org/10.32604/cmc.2021.018993).
- [35] R. B. Maher and O. A. Nasr, "DropStore: A secure backup system using multi-cloud and fog computing," *IEEE Access*, vol. 9, pp. 71318–71327, 2021, doi: [10.1109/ACCESS.2021.3078887](https://doi.org/10.1109/ACCESS.2021.3078887).
- [36] M. Dickinson, S. Debroy, P. Calyam, S. Valluripally, Y. Zhang, R. B. Antequera, T. Joshi, T. White, and D. Xu, "Multi-cloud performance and security driven federated workflow management," *IEEE Trans. Cloud Comput.*, vol. 9, no. 1, pp. 240–257, Jan. 2021, doi: [10.1109/TCC.2018.2849699](https://doi.org/10.1109/TCC.2018.2849699).
- [37] S. De Capitani di Vimercati, S. Foresti, G. Livraga, V. Piuri, and P. Samarati, "Security-aware data allocation in multicloud scenarios," *IEEE Trans. Dependable Secure Comput.*, vol. 18, no. 5, pp. 2456–2468, Sep. 2021, doi: [10.1109/TDSC.2019.2953068](https://doi.org/10.1109/TDSC.2019.2953068).



DRAJAT CAHYA DININGRAT received the bachelor's degree in informatics engineering from Pasundan University, Bandung, in 2018. He is currently pursuing the master's degree with the School of Electrical Engineering and Informatics, Bandung Institute of Technology (ITB). He holds several professional certifications, such as Certified Secure Computer User from EC-Council and Microsoft Certified Security Operation Analyst Associate. His current research interests include cybersecurity and cloud computing.



SUHARDI (Member, IEEE) received the bachelor's and master's degrees in electrical engineering from Bandung Institute of Technology (ITB), Indonesia, in 1988 and 1992, respectively, and the Dr.-Ing. degree in electrical engineering from the Technical University of Berlin, Germany, in 1997. He is currently a Professor with the School of Electrical Engineering and Informatics, ITB. He is also the Head of the Cyber Security Center, ITB. His research interests include cyber security, privacy and forensics, service computing, digital service transformation, and blockchain.



BUDI RAHARDJO (Member, IEEE) received the B.S. degree in electrical engineering from Bandung Institute of Technology (ITB), Indonesia, in 1986, and the M.S. and Ph.D. degrees in computer science from the University of Manitoba, Winnipeg, MB, Canada, in 1990 and 1997, respectively. He is currently a Researcher and a Lecturer with the School of Electrical Engineering and Informatics, ITB. His current research interests include computer security and cryptography. He is the Founder and the Chairperson of Indonesia Computer Emergency Response Team (ID-CERT). He is one of the founders of Asia-Pacific Computer Emergency Response Team (AP-CERT).

...